Omsk2406

Qu 1

$[7]^{-1} \odot [3]$ i $\mathbb{Z}_{15}$

$7x \equiv 1 \pmod{15}$

$15 = 2 \cdot 7 + 1$

$1 = 15 - 2 \cdot 7$

$-2 \equiv 13 \pmod{15}$

Den modulära inversen av 7:

$[7]^{-1} = [13]$ i $\mathbb{Z}_{15}$

$[13] \cdot [3] = 13 \cdot 3 = 39$

$39 \bmod 15 = 9$

$\boxed{\text{Därmed, } [7]^{-1} \odot [3] = [9] \text{ i } \mathbb{Z}_{15}}$